

Pruning Vision Transformers Using Inter-Layer Similarity Measures and MLP-Based Patch Skipping

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Abstract—Vision Transformers (ViTs) have demonstrated state-of-the-art performance in various computer vision tasks. However, their high computational cost, primarily due to processing many image patches and the self-attention mechanism, poses challenges for deployment on resource-constrained devices. We conduct a preliminary analysis of cosine similarity between patch embeddings across layers in ViTs, revealing patterns of redundancy. Based on these insights, we propose a novel MLP-based pruning approach that dynamically identifies patches for skipping based on cosine similarity between consecutive layers, indicating minimal feature evolution. A key aspect of our methodology is an alternating training strategy, where the patch-pruning MLP and the ViT are iteratively fine-tuned. Our method allows for the reintroduction of skipped (idle) patches in subsequent layers, mitigating information loss. We evaluate our approach on CIFAR-100 and a subset of ILSVRC (ImageNet-1K), presenting results, ablation studies, and discussing computational savings. The challenges encountered, including handling variable patch counts, modifying attention mechanisms, and the sensitivity of similarity thresholds, are detailed. Furthermore, we explore the potential of extending this methodology to dense prediction tasks like semantic segmentation.

Index Terms—Vision Transformers, Pruning, Patch Pruning, Token Pruning, Cosine Similarity, MLP, Computational Efficiency, Semantic Segmentation, Alternating Training.

I. INTRODUCTION

Vision Transformers (ViTs) achieve strong performance but require high computational resources due to processing many image patches. To address this, patch pruning—removing less informative patches—reduces the number of tokens, leading to lower computational cost without altering the model architecture. Efficient pruning techniques aim to alleviate this computational burden by reducing the number of parameters or computations in ViTs without significant degradation in performance.

We focus on exploring existing pruning strategies for ViTs and proposing an innovative method to optimize ViTs for tasks such as classification and, potentially, semantic segmentation. The core problem we address is identifying redundant patches within ViTs and skipping their computation to save resources. While this is explored in the context of classification, we also consider its applicability to semantic segmentation, where every patch can be critical for dense prediction. Our objective is to develop and test a novel pruning approach that utilizes similarity measures such as cosine similarity and Euclidean distance to identify and skip redundant patches, coupled with

an alternating training strategy for the pruning module and the ViT, with an initial focus on classification tasks and a view towards future extension to semantic segmentation.

II. RELATED WORK

Several pruning and token reduction methods have been proposed for Vision Transformers. Table I provides a summary of some notable approaches.

A. Specific Pruning Approaches from Literature

Patch Slimming identifies effective patches in the final layer and uses this information to guide pruning in earlier layers. It measures the impact of patches on output features and discards those with minimal contribution, achieving up to a 45% reduction in FLOPs for ViT-Ti with only a 0.2% drop in top-1 accuracy on ImageNet.

XPruner introduces a pruning framework centered on the explainability of the pruning criterion. It quantifies the contribution of each prunable unit to predicting specific target classes using an explainability-aware mask learned end-to-end. This method dynamically determines layer-wise pruning thresholds to preserve informative units and optimize pruning rates adaptively.

IARED² (Interpretability-Aware Redundancy Reduction) identifies and removes uncorrelated patches dynamically across different stages. It employs a policy network (multi-head interpreter) to decide which patches are uninformative and discard them. This approach is model-agnostic and task-agnostic, achieving up to 1.4x speed-up with less than 0.7% accuracy drop for DeiT and TimeSformer models.

Dynamic Token Pruning in Plain Vision Transformers for Semantic Segmentation (DToP) [5] proposes a mechanism that adaptively halts (“exits”) easy-to-predict tokens in early layers of a ViT and only forwards the remaining hard tokens to deeper layers. By leveraging the model’s existing auxiliary-loss heads to gauge token “difficulty,” DToP prunes easy tokens on the fly—while retaining a small, fixed number of high-confidence tokens per class to preserve context—and thus reduces computation by 20–35% without modifying the backbone architecture or harming accuracy.

Adaptive Token Sampling For Efficient Vision Transformers (ATS) [2] inserts a differentiable, parameter-free scoring-and-sampling layer into existing ViT blocks. At each

TABLE I
SUMMARY OF PRUNING AND TOKEN REDUCTION METHODS IN VISION TRANSFORMERS.

Paper/Method	Type	Dataset	Base Transformer	Base Accuracy Top1 — % Drop	Base GFLOPs — % Drop
Adaptive Token Sampling For Efficient Vision Transformers [2]	token pruning	Imagenet	DeiT-S	79.8 — 0.1%	4.6 — 36.9%
			PS-ViT-B	82.3 — 0.1%	8.8 — 36.36%
DynamicViT: Efficient Vision Transformers with Dynamic Token Sparsification	token sparsification	Imagenet	DeiT-S	79.8 — 0.5%	4.6 — 37%
			LV-ViT-S	83.3 — 0.3%	6.6 — 31%
			LV-ViT-M	84 — 0.2%	12.6 — 33%
AdaViT: Adaptive Tokens for Efficient Vision Transformer	adaptive tokens	Imagenet	DEiT-S → A-ViT-S	79.8 — 0.3%	4.6 — 38%
			DeiT-T → A-ViT-T	71.3 — 0.3%	1.2 — 33%
Token Merging: Your ViT but faster	token merging	Imagenet	DeiT-S	79.4 — 0.5%	2.7 — 41.3%
			DeiT-S (AugReg)	79.3 — 0.6%	2.7 — 41.3%
Patch Slimming	token pruning	ImageNet	DeiT-S → PS-ViT	79.4 — 0.5%	2.6 — 43.6%
			DeiT-S → DPS-ViT	79.5 — 0.4%	2.4 — 47.8%
IdleViT	token idling	ImageNet	DeiT-S	79.8 — 0.1% at keep ratio of 0.8	3.5(GMacs) — 24%
			DeiT-S	79.3 — 0.6% at keep ratio of 0.6	2.7(GMacs) — 42%
PPT: Token Pruning and Pooling for Efficient Vision Transformers	token pruning + token merging	ImageNet	DeiT-S	79.5 — 0.4% (off-the-shelf)	2.9 (↓ 37.0%)
			DeiT-S	79.8 — 0.1%	2.9 (↓ 37.0%)

insertion point, ATS uses the classification token’s self-attention weights to assign an importance score to every patch token. It then stochastically samples a subset of tokens—retaining more tokens for complex images and fewer for simpler ones—via inverse-transform sampling.

B. Cosine Similarity Insights and Related Works

Deeper ViT layers often produce embeddings for different patches that are highly similar, indicating increased cosine similarity as layers deepen. High cosine similarity among patch embeddings can lead to a loss of discriminative features. Some works introduce novel loss functions to explicitly reduce cosine similarity and encourage diversity.

Anti-Oversmoothing in Deep Vision Transformers identifies attention collapse or patch uniformity as a limitation in deeper ViTs, where embeddings across different patches become nearly identical. Analysis in the Fourier spectrum shows that the low-pass filtering nature of self-attention suppresses high-frequency components, causing patches to converge. This results in the loss of finer, localized details and is evidenced by increased cosine similarity between patch embeddings.

Skip Attention (SKIPAT) identifies high redundancy in self-attention computations across layers in ViTs. SKIPAT reuses self-attention computations from earlier layers to approximate results in subsequent layers, introducing a lightweight parametric function for this purpose.

No Token Left Behind (IdleViT) highlights inefficiencies in early-stage token pruning, which can permanently lose crucial image information. It proposes a dynamic token idling

mechanism that keeps unselected tokens “idle” and passes them to subsequent layers. This allows idle tokens to re-enter computations later, mitigating the negative impact of early-stage pruning. Our approach shares a similar motivation to avoid permanent information loss, though we use cosine similarity for patch skipping/idling.

III. METHODOLOGY

A. Datasets

1) **CIFAR-100**: The CIFAR-100 dataset consists of 60,000 color images of size 32×32 pixels, evenly divided into 100 fine-grained classes (600 images per class). These 100 classes are further grouped into 20 superclasses, providing both “fine” labels (the specific class) and “coarse” labels (the superclass) for each image. There are 50,000 training images and 10,000 test images, with 500 training and 100 test samples per class. This dataset was used for initial experiments and analysis.

2) **ILSVRC (ImageNet-1K) Subset**: The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) “ImageNet-1K” benchmark comprises 1,281,167 training images, 50,000 validation images, and 100,000 test images spanning 1,000 object categories. Due to GPU memory and time constraints, we did not use the full 1,000-class set; instead, we randomly sampled a smaller subset of classes (and a proportional fraction of images per class) to create a lightweight benchmark that still captures diverse object categories. This subset enables faster iteration while preserving the challenge of large-scale, fine-grained classification.

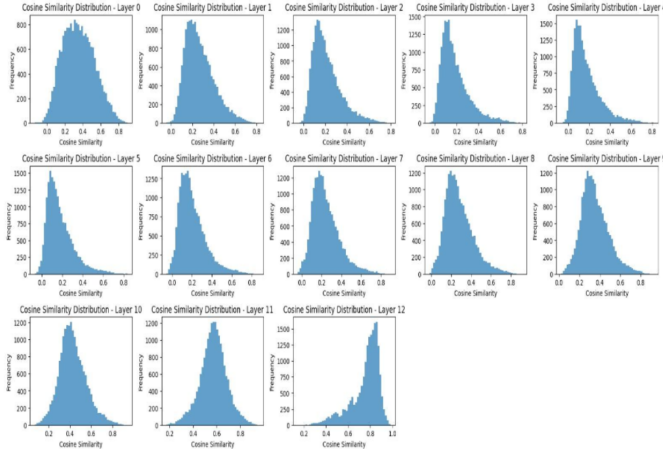


Fig. 1. Layer-wise cosine similarity distribution in CIFAR-100. Lower layers (e.g., Layer 0) show broader distributions, while deeper layers (e.g., Layer 12) show narrower distributions centered at higher similarity values.

B. Preliminary Cosine Similarity Analysis

1) **Experimental Setup:** To investigate cosine similarity trends and identify redundancy patterns in ViTs, we performed an initial analysis on CIFAR-100.

• Metrics/Plots:

- Distribution of cosine similarity of every combination of patches at each layer (across all images).
- Average cosine similarity of every combination of patches at each layers (across all images).
- Cosine similarity between same-positioned patches across adjacent layers (our motivation).

2) **Observations:** The layer-wise cosine similarity (similarity between all patches within the same layer) showed that lower layers have broader distributions, indicating diverse patch representations. In deeper layers, the similarity distribution narrows and its center shifts towards higher cosine similarity values, highlighting more uniform processing of patch embeddings. This is illustrated in Figure 1.

The average cosine similarity across all images for each layer is higher in the later layers of the ViT. A noticeable dip occurs in the middle layers, reflecting more diverse and decorrelated feature representations. The sharp rise in similarity towards the final layers implies convergence on a more unified or global representation. These findings validate observations from existing literature (Figure 2).

Critically for our approach, the cosine similarity between same-positioned patches across adjacent layers increases as we move to deeper layers (highest between layers 6-10). This indicates that patch representations stabilize and do not update significantly in these deeper layers. Lower layers exhibit more dynamic feature extraction, reflected by darker regions (lower similarity) in the heatmap (Figure 3).

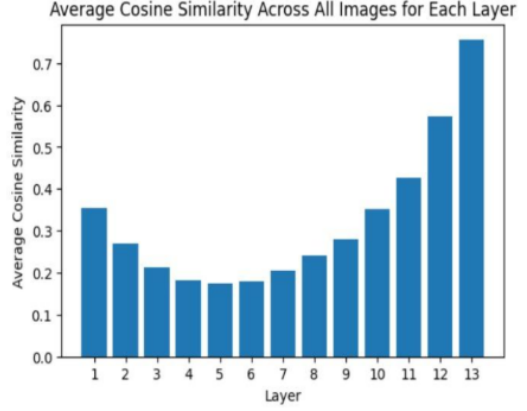


Fig. 2. Layer-wise average cosine similarity on CIFAR-100. Similarity dips in middle layers and rises sharply in later layers.

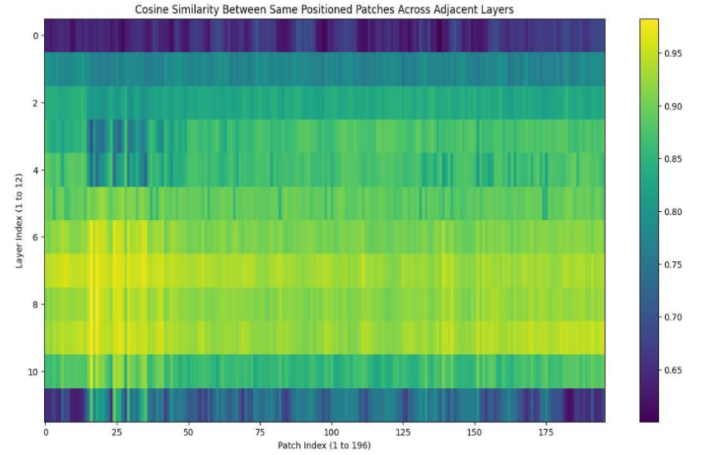


Fig. 3. Cosine similarity between same-positioned patches across adjacent layers in CIFAR-100. Yellow regions indicate high similarity, more prevalent in deeper layers (e.g., Layer Index 6-10).

C. Core Idea and Motivation for Proposed Approach

High cosine similarity between similarly positioned patches across consecutive layers suggests these patches are not learning new features at that stage, marking them as redundant. Pruning such patches could lead to significant computational savings without adversely affecting model performance. Our approach introduces comparing cosine similarity between similarly positioned patches at layer i and layer $(i + 1)$ to identify patches that remain static. This focus on skipping patches based on inter-layer cosine similarity metrics, and training a dedicated MLP for this task via an alternating scheme, forms the novel aspect of our method.

D. Alternating Training and Inference Workflow

To ensure effective optimization of both the lightweight MLP and the Vision Transformer (ViT) components, we propose a two-phase alternating training strategy. Simultaneous training often results in a loss imbalance, leading to overfitting

of one component. Our approach addresses this by decoupling the early-stage training and introducing a controlled alternation between modules.

Warm-up Phase: We begin by training only the MLP for $n = 10$ epochs. This helps it establish a robust initial representation.

Joint Alternating Phase: Post warm-up, we alternate between training the MLP and the ViT. This alternating pattern helps mitigate overfitting and ensures balanced learning across both modules.

The complete training and inference process is summarized below:

Algorithm 1 Alternating Training and Inference Workflow

- 1: **Training Phase**
 - 2: Compute scaled cosine similarity between corresponding patches across layers i and $(i + 1)$
 - 3: Generate labels based on `cosine_threshold`:
 - 4: **if** scaled cosine similarity $>$ `cosine_threshold` **then**
 - 5: Label = 0 (Skip Patch)
 - 6: **else**
 - 7: Label = 1 (Keep Patch)
 - 8: **end if**
 - 9: Use the lightweight MLP to predict a pruning score in $[0, 1]$, inversely proportional to the scaled cosine similarity
 - 10: Compute Cross-Entropy loss between generated labels and predicted scores
 - 11: Backpropagate the loss to update MLP weights
 - 12: Alternately update ViT and MLP during joint training phase
 - 13: **Note:** Skipped patches may be reintroduced in subsequent layers
 - 14: **Inference Phase**
 - 15: For each patch, compute pruning score using the MLP
 - 16: Apply `score_threshold` for binary decision:
 - 17: **if** MLP score $>$ `score_threshold` **then**
 - 18: Keep Patch (Label = 1)
 - 19: **else**
 - 20: Skip Patch (Label = 0)
 - 21: **end if**
 - 22: Compute scaled cosine similarity and ground truth labels for each patch
 - 23: Compare predicted labels with ground truth to calculate accuracy = 0
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E. Characteristics of the Proposed Approach

- **Dynamic Pruning of Unevolving Patches:** High cosine similarity identifies redundant patches that are not evolving in terms of features. Our MLP-based mechanism dynamically evaluates patch relevance at each layer.
- **Reintroduction of Idle Patches:** Unlike methods that permanently drop tokens/patches, our approach allows idle patches to be passed to subsequent layers for reevaluation, preserving information.

IV. RESULTS AND DISCUSSION

Experiments were conducted on CIFAR-100. The base ViT accuracy without pruning was 89.8%.

A. Performance with Pruning

Table II shows the ViT accuracy and average number of patches pruned at different cosine similarity thresholds from the initial experiments.

TABLE II
ACCURACY OF ViT AND AVERAGE PATCHES PRUNED AT DIFFERENT THRESHOLDS (10 EPOCHS)

Threshold	# Patches Pruned	FLOPs decrease %	Accuracy
0.95	40	~ 20%	85.2%
0.9	68	~ 35%	77%

Table III presents results using a modified attention mechanism.

TABLE III
ACCURACY OF ViT AND AVERAGE PATCHES PRUNED AT THRESHOLD = 0.95 USING MODIFIED ATTENTION (10 EPOCHS)

Threshold	# Patches Pruned	FLOPs decrease %	Accuracy
0.95	40	~ 16%	85.4%

The “Average patches skipped” refers to the number of patches skipped out of 196 on average.

B. MLP Accuracy and Skip Ratio Across Layers

Table IV details the MLP’s accuracy in predicting skippable patches and the average number of patches skipped per layer.

TABLE IV
MLP ACCURACY AND SKIP RATIO ACROSS DIFFERENT LAYERS OF ViT

Layer	MLP Accuracy	Average Patches Skipped
1	1.0000	0
2	1.0000	0
3	0.9997	0.0145
4	0.9127	8.1373
5	0.8542	19.672
6	0.7684	45.654
7	0.8108	149.31
8	0.8726	173.68
9	0.8310	153.91
10	0.8045	157.21
11	0.9872	7.4308
12	0.9922	4.5522

The MLP accuracy remains high in initial and later layers but decreases in the middle layers (6-10). Conversely, the average number of patches skipped increases in these middle layers. This might suggest that while pruning is more active in these layers, the MLP’s ability to precisely identify skippable patches is reduced. This observation is consistent with the earlier finding that same-positioned patches show high cosine similarity (stabilization) particularly in layers 6-10.

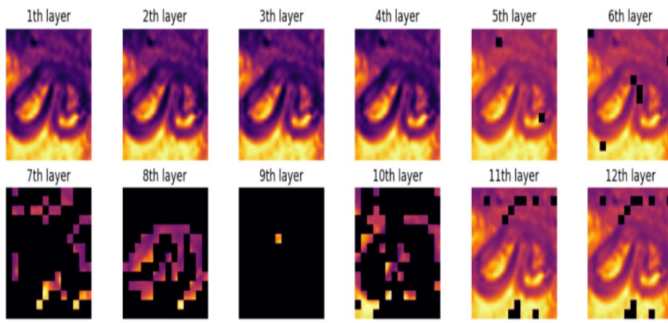


Fig. 4. Visualization of Skipped Patches (MLP mask) at each layer for CIFAR-100 ‘snake’ class (Illustrative). Black areas indicate skipped patches.

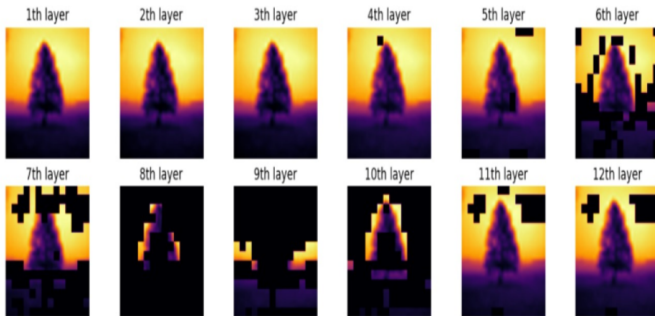


Fig. 5. Visualization of Skipped Patches (MLP mask) at each layer for CIFAR-100 ‘tree’ class (Illustrative). Black areas indicate skipped patches.

C. Visualization of Skipped Patches

Visualizations (Figure 4 and 5) show the MLP mask at each layer for specific classes from CIFAR-100.

D. Observations from Visualizations

- **Progressive Skipping:** Skipped patches (black-out areas) gradually increase in deeper layers, with significant skipping starting around the 7th layer. This aligns with the increasing cosine similarity observed in deeper layers.
- **Early Layers Retain Information:** In initial layers (1-6), most patches are retained, suggesting these layers focus on extracting general features. Aggressive pruning here might harm performance.
- **Effective Skipping in Mid-Deeper Layers:** At layers 8 and 10, the pruning strategy appears to correctly black out irrelevant background patches while retaining critical foreground ones.

E. Ablation Studies

To evaluate the effectiveness of different architectural choices for patch classification, we experimented with various models, including simple MLPs, CNNs, and transformer-based configurations. Each model was assessed based on classification accuracy, MLP accuracy, and the percentage of skipped patches. The results, summarized in Table V,

highlight the trade-offs between computational efficiency and performance. Notably, architectures incorporating both CLS and an additional token demonstrated competitive performance while maintaining a relatively high skip ratio.

TABLE V
COMPARISON OF PROPOSED AND BASELINE ARCHITECTURES FROM ABLATION STUDIES. ALL MODELS ARE TRAINED FOR 5 EPOCHS UNLESS SPECIFIED.

Method	Accuracy (%)	MLP Accuracy (%)	Skip Ratio (%)
Simple MLP [†]	51.00	90.00	–
ViT Fine-tuning [‡]	82.00	85.00	–
UpdateNet + MLP [§]	25.20	71.00	28.06
CNN Model [¶]	88.40	75.30	8.16
All Patches	59.00	87.89	38.00
Focal Loss ^{††}	85.86	82.13	18.88
Shared MLP ^{‡‡}	49.83	52.70	62.76
CLS + Token Only ^{§§}	84.66	84.62	27.55

Notes:

- [†] Trained using a simple MLP with learning rate of 10^{-3}
- [‡] ViT model initialized with the MLP and fine-tuned
- [§] UpdateNet configuration: [CLS+Patch, 384, 768]; MLP: [768, 64, 1]
- [¶] CNN with progressive downsampling ($56 \times 56 \rightarrow 14 \times 14$)
- ^{||} Patch-wise classification using encoder: [768, 128, 64, 16]
- ^{††} Trained with focal loss
- ^{‡‡} Shared MLP across layers
- ^{§§} Architecture utilizes only CLS token and one additional token

V. ANALYSIS OF PATCH SKIPPING BEHAVIOR AND MLP PERFORMANCE

To gain deeper insights into the proposed patch skipping mechanism and the performance of the Multi-Layer Perceptron (MLP) in identifying skippable patches, a detailed error analysis was conducted. This involved examining the frequency of patch skipping by the MLP’s binary decision, comparing it against the ground truth derived from cosine similarity thresholds, visualizing misclassifications, and inspecting qualitative examples.

A. Patch Skip Frequency Analysis

1) *MLP-Predicted Skip Frequencies:* The frequency at which patches were skipped by the MLP’s binary decision across different layers of the Vision Transformer (ViT) was analyzed over the entire validation set. The skip frequency for each patch position is scaled to 1, where a value of 0.5 indicates that the patch at that specific position was skipped for 50% of the images in the validation set.

Figures 6 and 7 illustrate these MLP-predicted skip frequencies for eight different layers.

Observations for MLP-Predicted Skips:

- **Initial Layers (e.g., Layers 0-3):** As depicted in Figure 6, the MLP skips very few patches in the initial layers. The heatmaps show predominantly low skip frequencies (values close to 0, indicated by darker blue in the provided PDF), suggesting that the MLP deems most patches as important for processing in these early stages.

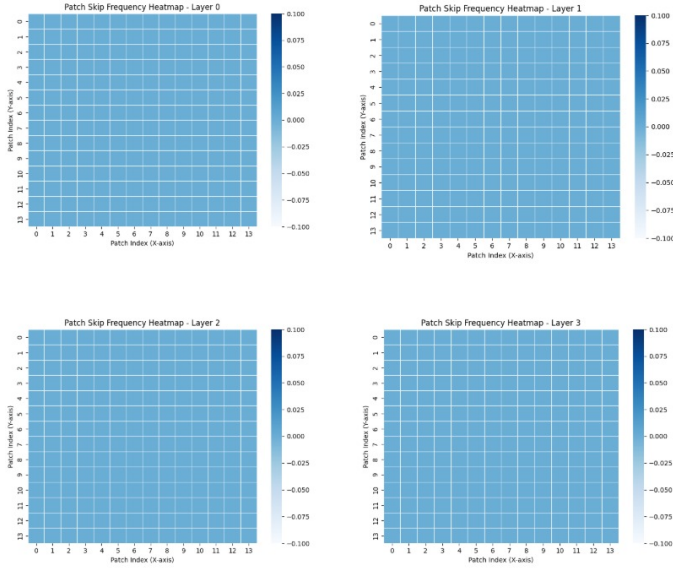


Fig. 6. MLP-Predicted Patch Skip Frequency Heatmaps for Early to Mid Layers (e.g., Layers 0, 1, 2, 3).

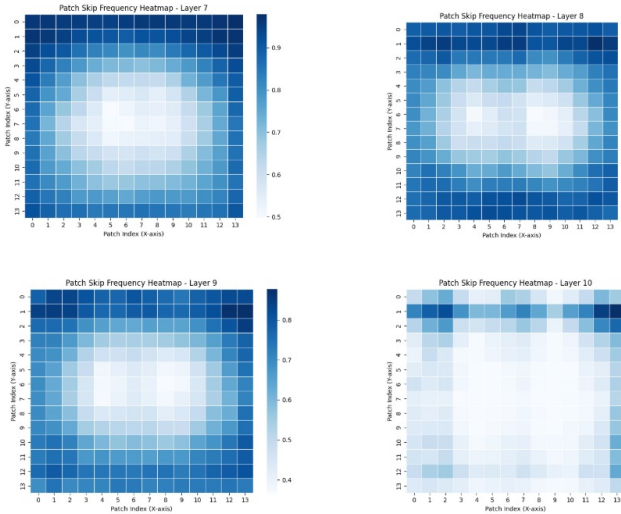


Fig. 7. MLP-Predicted Patch Skip Frequency Heatmaps for Mid to Later Layers (e.g., Layers 7, 8, 9, 10).

This aligns with the expectation that initial layers capture fundamental visual features across the image.

- **Later Layers (e.g., Layers 7-10):** In contrast, Figure 7 shows a significant increase in patch skipping in later layers. Patches, particularly in the central regions, exhibit higher skip frequencies (lighter areas in the heatmaps from the PDF). Layer 8, for instance, shows a very high frequency of skipping for many patches, and Layer 10 indicates that a large majority of patches are skipped most of the time. This suggests the MLP identifies substantial redundancy or consolidates information into fewer patches in deeper stages of the ViT.

- **Spatial Pattern:** In layers with significant skipping (e.g., 7, 8, 9), there's often a tendency for peripheral patches to be kept (lower skip frequency) more often than central patches, although Layer 10 shows widespread skipping. This might imply that edge information remains relevant longer or that central features become redundant faster.

2) *Ground Truth Skip Frequencies:* For comparison, the patch skip frequencies were also determined based on the ground truth mechanism, i.e., by applying a threshold to the cosine similarity values between patch representations in consecutive layers. These represent the "ideal" skipping pattern our MLP aims to learn.

Figures 8 and 9 show these ground truth skip frequencies.

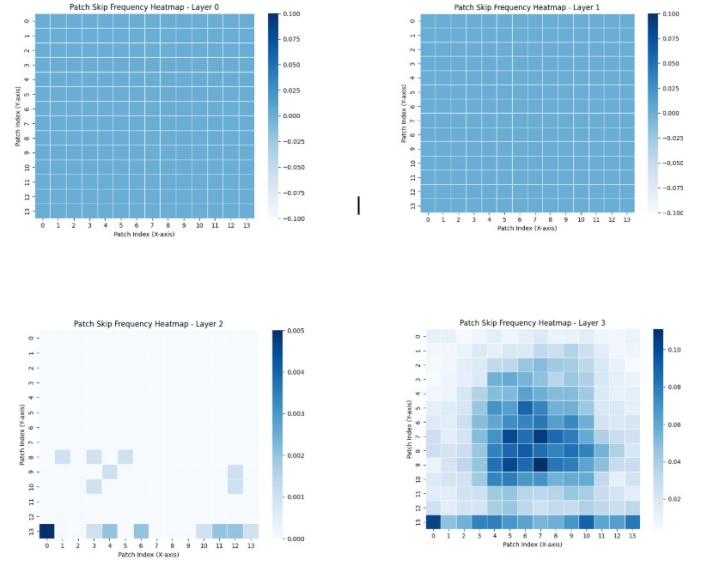


Fig. 8. Ground Truth Patch Skip Frequency Heatmaps for Early to Mid Layers (e.g., Layers 0, 1, 2, 3).

Observations for Ground Truth Skips:

- **Initial Layers (e.g., Layers 0-2):** Figure 8 shows that, according to the ground truth (cosine similarity thresholding), there is virtually no skipping in Layers 0 and 1, and very minimal, sparse skipping in Layer 2. This is because patch representations are expected to change significantly in early layers as basic features are extracted, resulting in low inter-layer cosine similarity.
- **Transition Layer (e.g., Layer 3):** Layer 3 begins to show a more structured pattern of potential skipping according to ground truth, with certain central patches exhibiting higher skip frequencies. This marks the beginning of feature stabilization for some patches.
- **Later Layers (e.g., Layers 7-10):** Similar to the MLP predictions, Figure 9 indicates that the ground truth supports more aggressive skipping in later layers. The patterns of higher skip frequencies in central regions are also evident, particularly in Layers 7, 8, and 9. Layer 10 also shows widespread high skip frequencies.

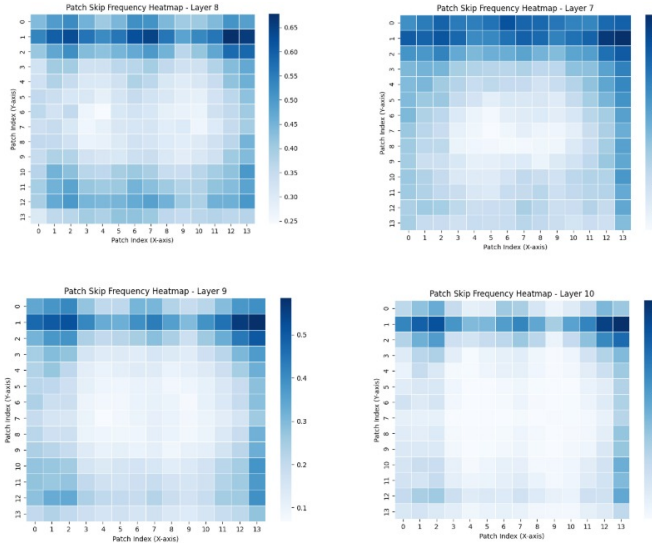


Fig. 9. Ground Truth Patch Skip Frequency Heatmaps for Mid to Later Layers (e.g., Layers 7, 8, 9, 10).

Cross-Observations (MLP vs. Ground Truth Skip Frequencies):

- **Initial Layers Agreement:** Both the MLP and the ground truth indicate minimal to no skipping in the very early layers (e.g., Layers 0, 1). This is a positive sign, as it suggests the MLP correctly learns not to discard patches when their feature representations are still highly dynamic and informative.
- **Later Layers General Trend Agreement:** In later layers (7-10), both methods agree on the general trend of increased skipping and often highlight similar spatial regions (e.g., central patches) as candidates for skipping. For example, comparing MLP Layer 7 (Fig. 7) with Ground Truth Layer 7 (Fig. 9), both show a tendency to skip central patches more frequently.
- **Differences in Aggressiveness and Specificity:** While general trends often align, there can be differences in the exact skip frequencies and the “crispness” of the skipped regions.
 - The MLP’s predicted skip patterns sometimes appear smoother or more diffuse than the ground truth patterns, which can occasionally be more binary (either skip with high frequency or don’t skip).
 - In some layers, the MLP might be more or less aggressive than the ground truth. For example, MLP Layer 8 (Fig. 7) appears to skip a very large central block with high frequency, which is also seen in Ground Truth Layer 8 (Fig. 9), but the exact boundaries and intensity might differ slightly. MLP Layer 10, in particular, shows extremely high skip frequencies across almost all patches, which is also largely reflected in the ground truth for Layer 10.
- **Learning Success:** The concordance, especially in the

early layers (low skip) and the general pattern in later layers (high skip in central regions), suggests the MLP is reasonably successful in learning the underlying feature stability criteria captured by cosine similarity.

B. Patch Misclassification Analysis

To understand where the MLP fails to correctly predict whether a patch should be skipped or kept (compared to the ground truth similarity thresholding), patch misclassification heatmaps were generated. These heatmaps highlight regions where the MLP’s decision (skip/keep) differs from the ground truth decision. This analysis was focused on Layers 7-10, where significant skipping occurs.

Figure 10 shows the misclassification rates for these layers.

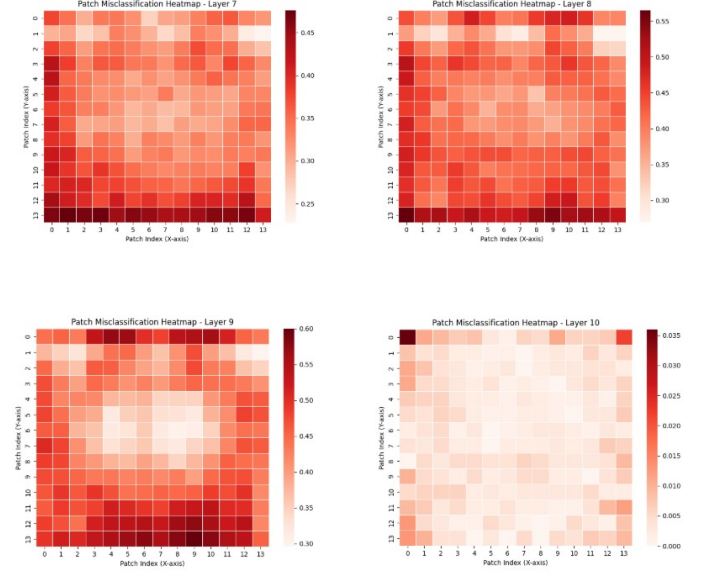


Fig. 10. Patch Misclassification Heatmaps for Layers 7, 8, 9, and 10. Higher values indicate higher misclassification rates for those patch positions.

Observations for Misclassification Heatmaps (Layers 7-10):

- **Layer 7:** Misclassifications appear to be spread out but are somewhat more pronounced towards the periphery and corners. Some central patches also show moderate misclassification.
- **Layer 8:** This layer exhibits relatively higher misclassification rates compared to Layer 7, especially in bands around the central region that the MLP aggressively skips. This could mean the MLP sometimes skips patches that the ground truth would keep, or vice-versa, in these transitional zones.
- **Layer 9:** Misclassification rates in Layer 9 are generally lower in the very central region (where both MLP and GT agree on high skip rates) but are noticeable in the surrounding patches. The pattern suggests that the MLP might struggle with the exact boundary of the “highly skippable” central zone defined by the ground truth.

- **Layer 10:** Layer 10 shows very low misclassification rates overall (darker regions in the heatmap from the PDF, indicating values close to 0). This is likely because both the MLP and the ground truth advocate for skipping almost all patches with very high frequency in this layer, leading to high agreement. The few misclassifications appear at the extreme edges.
- **General Trend:** The misclassification patterns suggest that the MLP learns the broad regions of “to skip” and “to keep” patches but might struggle more at the boundaries of these regions or with patches whose similarity scores are close to the decision threshold. Layers where the skipping behavior is more mixed (not uniformly skip or keep by ground truth) tend to show higher, more dispersed misclassification.

C. Qualitative Analysis of Skipping Behavior

To provide a more intuitive understanding, Figure 11 shows two specific instances from the validation set. For each instance, it compares the patches skipped by the MLP (left column of the pair for each layer/instance) versus the patches that were supposed to be skipped according to the ground truth similarity thresholding (right column of the pair). The images likely depict a sequence of layers for these two instances.

Observations for Qualitative Examples :

- **Object Preservation:** A key observation across both instances and multiple layers is that both the MLP and the ground truth tend to avoid skipping patches that constitute the main object(s) in the image (e.g., the alarm clock). The red overlays (indicating skipped patches if we assume the convention from the presentation was blacked-out for skipped) or the preserved areas primarily focus on the object itself.
- **Background Skipping:** Conversely, patches corresponding to the background are more frequently identified for skipping by both the MLP and the ground truth. This is desirable, as background often contains less critical information for classification tasks.
- **MLP Aggressiveness:** In several displayed layers, the MLP appears to skip more patches than the ground truth. This is evident if the “skipped by MLP” (left column in pairs) shows more red or blacked-out regions compared to the “skipped by ground truth” (right column in pairs).
- **Complete Skip in Deep Layers (e.g., Layer 8):** One prominent example shows that for one of the instances, by Layer 8 (or a similarly deep layer shown), the MLP decision leads to nearly all patches being skipped (e.g., an almost entirely red or black image if skipping is visualized that way). The ground truth for that layer might also show extensive skipping but perhaps preserve a few more object-centric patches. If Layer 8 in the figure shows a fully skipped image by MLP, it reflects the high skip frequencies observed in Figure 7 for that layer.
- **Correspondence and Discrepancies:** While there’s a good general correspondence in targeting background over foreground, the exact set of skipped patches can

differ. The MLP might learn a slightly different, possibly more generalized, skipping strategy. For instance, if an object is small, the MLP might still prune aggressively around it, whereas the ground truth (based purely on local patch similarity) might be more conservative if those object-adjacent patches still show some dissimilarity to their counterparts in the next layer.

- **Progression Through Layers:** The visualization across layers would typically demonstrate how skipping becomes more aggressive in deeper layers, a trend consistent with the frequency heatmaps.

In summary, the error analysis indicates that the MLP learns meaningful patterns for patch skipping, generally aligning with ground truth criteria derived from cosine similarity. It correctly identifies that initial layers require minimal skipping and that later layers can afford more aggressive pruning, particularly of background or redundant central features. While some misclassifications and differences in skipping aggressiveness exist, the qualitative examples reinforce that the MLP prioritizes preserving object-relevant patches, which is crucial for maintaining task performance. The very high skip rates in the deepest layers (e.g., Layer 10, and notably Layer 8 by MLP) suggest a strong consolidation of information as processing nears the classification head.

VI. CHALLENGES FACED AND LIMITATIONS

Several challenges were encountered:

- **Variable Patch Count:** Pruning leads to a varying number of active patches per image within a batch.
- **Modified Attention Mechanism:** Unpruned patches might still need to attend to all patches.
- **Suboptimal Pruning Results (Initial):** Early methods reduced computational cost but didn’t achieve SOTA performance.
- **MLP Limitations:** A simple MLP operating on a single patch lacks broader context.
- **Sensitivity to Hyperparameters:** Our patch-pruning technique relies on similarity metrics requiring careful threshold tuning.

VII. RESULTS ON IMAGENET

Our baseline ViT model achieved 82.45% accuracy on ImageNet. With MLP accuracy at 100%, we achieved 80.16% accuracy after pruning 16.1% of tokens. With 86% MLP accuracy, we obtained 76.2% ViT accuracy at 15% pruning.

VIII. EXTENSION TO SEMANTIC SEGMENTATION

A. Motivation

Our pruning method showed intelligence by performing better than random pruning. Semantic segmentation benefits from global context, making our method potentially suited for high-resolution outputs.

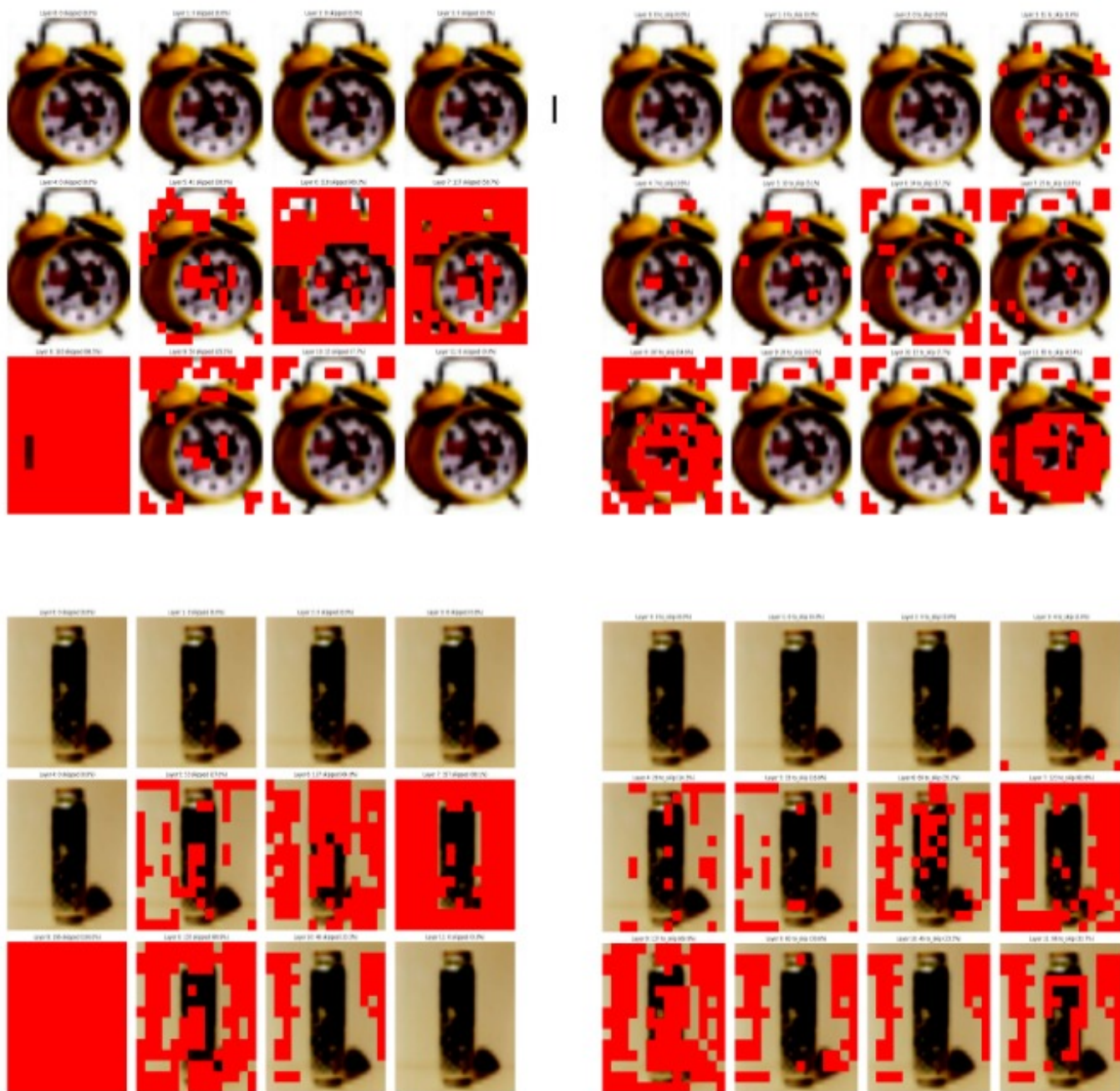


Fig. 11. Qualitative examples of patch skipping for two validation set instances. For each instance (e.g., alarm clock), a sequence of layers is shown. Within each layer’s visualization, the left sub-image typically shows patches skipped by the MLP decision (red overlays on skipped parts), and the right sub-image shows patches targeted for skipping by the ground truth (cosine similarity threshold).

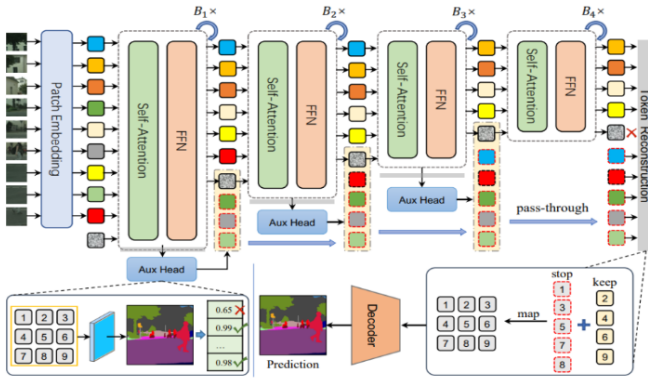


Fig. 12. DoViT Framework (Illustrative). Shows how tokens are divided.

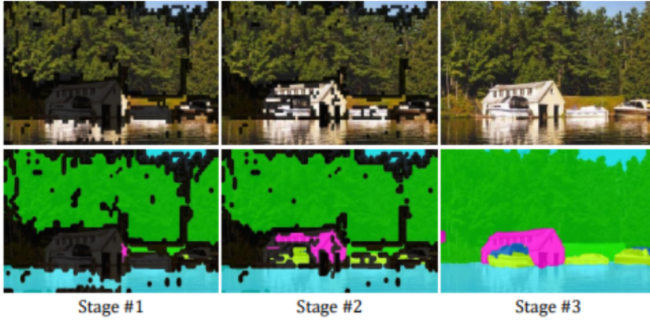


Fig. 13. Illustration of token difficulty levels. Shows stages of token pruning.

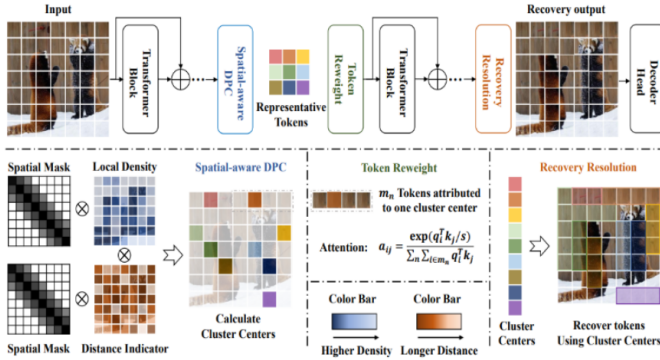


Fig. 14. AiluRus Framework (Illustrative). Shows token clustering.

B. Literature Review

Content-aware Token Sharing (CTS) merges tokens using class-agnostic policy networks.

Dynamic Token-Pass Vision Transformers (DoViT) dynamically stops processing "easy" tokens (Figure 12).

Dynamic Token Pruning (DToP) keeps k highest confidence tokens (Figure 13).

AiluRus uses spatial-aware clustering (Figure 14).

IX. FUTURE SCOPE

Future work could focus on:

- Enhance MLP predictions by investigating incorrect predictions
- Adapt the methodology to semantic segmentation
- Refine the pruning approach through further studies
- Investigate patch stability mechanisms
- Delve deeper into MLP learning patterns

X. CONCLUSION

In this work, we explored the feasibility of using similarity between patch embeddings across consecutive layers in Vision Transformers (ViTs) as a criterion for dynamically skipping redundant patches. Our preliminary analysis revealed that deeper layers exhibit higher cosine similarity between same-positioned patches, indicating potential redundancy and minimal feature evolution. Leveraging this observation, we proposed an MLP-based pruning approach coupled with an alternating training strategy to identify and skip such patches while allowing their reintroduction in subsequent layers.

However, our experiments highlighted notable limitations of this approach. While the method achieved computational savings by skipping patches—particularly in deeper layers where similarity was high—it also incurred a drop in model accuracy, indicating that similarity alone is insufficient for preserving critical features. Future improvements could integrate lightweight error correction (e.g., feature propagation), enhance the MLP with spatial or global context, or adapt thresholds dynamically. While challenges remain, the study opens doors for efficiency improvements in dense prediction tasks

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